

AraGenre 2026: A Hierarchical Definition-Guided Arabic Genre Classification Shared Task

Mo El-Haj^{1,2}, Saad Ezzini³, Shadi Abudalfa^{4,5},
Mustafa Jarrar⁶, Nguyen Minh Chi¹, Nguyen Minh Quan¹

¹VinUniversity, Vietnam

²Lancaster University, UK

³King Fahd University of Petroleum and Minerals, Saudi Arabia

⁴Onaizah Colleges, Saudi Arabia

⁵University College of Applied Sciences, Palestine

⁶Hamad Bin Khalifa University, Qatar

m.el-haj@lancaster.ac.uk

Abstract

AraGenre 2026 is a shared task on hierarchical, definition-guided Arabic genre classification under limited supervision and substantial distribution shift. Systems assign each text a broad label and a specific label using an inventory of six broad branches and 74 defined final genres. The final benchmark contains 27,972 source-derived Arabic segments, spanning Modern Standard Arabic, Classical Arabic, and dialectal material, while the released labelled examples are small and primarily synthetic. Final specific classes are absent from the released labelled data, but their English names and definitions are provided. The task received 46 registrations and 373 submissions, with 17 ranked final entries and 11 system description papers. Thaka achieved the highest official Hierarchical Macro F1 at 73.52, followed by HP at 71.69 and NAMAA-1 at 70.13. Broad classification was consistently stronger than specific classification. The documented systems frequently combine label semantics with multilingual retrieval, LLM judging, pseudo-labelling, reranking, or calibration. We clarify the resource-driven taxonomy and inherited gold labels, distinguish combined distribution shift from isolated unseen-label effects, and provide systematic validation of the final evaluation data.

1 Introduction

Genre characterises recurrent communicative purposes and discourse organisation, rather than subject matter alone. Functional accounts make these properties operational for corpus description and text classification (Sharoff, 2018; Ferhat et al., 2024). For Arabic, classification must also accommodate variation across Modern Standard Arabic (MSA), Classical Arabic, and dialects, including written code-switching and shared lexical forms (El-Haj et al., 2018; El-Haj, 2026). Genre, topic, regis-

ter, and linguistic variety can consequently provide overlapping signals without being interchangeable targets.

AraGenre 2026 examines this problem in a definition-guided setting. A conventional classifier learns associations between a fixed label inventory and labelled examples. AraGenre instead supplies natural-language label descriptions and introduces final specific classes that have no examples in the released labelled data. Systems must align Arabic passages with English descriptions while transferring from a small, largely synthetic development setting to a much larger source-derived evaluation collection. The task therefore tests a combination of label-semantic transfer and distributional robustness, not either factor in isolation.

The benchmark consolidates existing Arabic resources. Its gold specific labels are inherited from source categories or corpus-defined document types; the organisers contribute a common upper-level grouping, English definitions, task splits, baselines, and evaluation infrastructure. The resulting taxonomy is pragmatic and resource-driven. Some leaves distinguish document types, while others refine them by topic, occupation, educational level, or linguistic variety. This distinction is important both for interpreting the task and for understanding possible source and domain shortcuts.

This overview describes the taxonomy and construction process, presents the official results for 17 final leaderboard entries and the 11 accompanying system papers, and analyses the strategies used by documented systems. It also provides additional validation and documentation of the final evaluation data. Appendix A lists every defined specific label, its broad parent, and its observed support. Appendices B–D document verified data properties and provenance boundaries, define a prediction-

consistency diagnostic, and outline controlled evaluations for future editions. These structural checks complement rather than replace semantic annotation and source-level documentation.

2 Background and Motivation

Functional approaches to web and Arabic text classification recognise that communicative behaviour is not reducible to topic (Sharoff, 2018; Ferhat et al., 2024). The Arabic Functional Text Dimensions Corpus contains 3,400 documents across 17 categories (Ferhat et al., 2024). AraGenre differs in providing a larger final label inventory whose members are not observed as labelled classes during development. Its labels nevertheless retain distinctions inherited from heterogeneous source resources, so the benchmark should not be interpreted as an exhaustive or theoretically uniform ontology of Arabic genre.

Using label descriptions connects AraGenre to work on label-description training for zero-shot classification (Gao et al., 2023) and label augmentation for zero-shot hierarchical classification (Paletto et al., 2024). English definitions give systems a shared semantic interface to otherwise unseen Arabic categories. This motivates both direct multilingual retrieval and models that compare a passage against candidate descriptions.

Here, *unseen* means absent as a labelled class from the task’s released training and development material. It does not establish that a genre, source text, or document has never appeared in a pretrained model’s data. Nor does it prohibit systems from creating synthetic examples or pseudo-labels for final categories. The definition-guided task setting and an individual system’s training regime must therefore be distinguished: some entries are direct zero-shot classifiers, whereas others adapt to the unlabelled final batch.

3 Task Design and Genre Taxonomy

3.1 Prediction problem

For each Arabic text x_i , a system returns a broad label $\hat{b}_i$ and a specific label $\hat{s}_i$. The released taxonomy defines a set $\mathcal{B}$ of six broad branches, a set $\mathcal{S}$ of 74 final specific genres, and a single parent mapping $p : \mathcal{S} \rightarrow \mathcal{B}$. A structurally consistent prediction satisfies $\hat{b}_i = p(\hat{s}_i)$. The official scorer evaluates broad and specific outputs separately and does not enforce this constraint.

3.2 Why these six broad branches?

The six branches are *Informative*, *Creative*, *Interactive*, *Learning*, *Legal*, and *Religious*. They provide a compact organisation of the available source categories rather than six mutually exclusive theoretical dimensions. Informative and Creative primarily foreground communicative function; Interactive foregrounds participant-directed communication; Learning groups pedagogical material; Legal and Religious retain institutional discourse traditions. Their rationale is coverage of the inherited inventory at a manageable upper level, not a claim that function, institution, topic, and register can always be separated.

The mapping is assigned at category level: all instances with the same specific genre receive the same broad parent. The released mapping makes the intended distinctions concrete. News about religion and descriptions of Islamic books are Informative, whereas scripture and tafsir are Religious. Legal-service vacancies and descriptions of sharia-law books are Informative, whereas constitutions, contracts, and resolutions are Legal. Education vacancies are Informative, while textbooks and student writing are Learning. These are properties of the released mapping, not a universal precedence rule for ambiguous Arabic texts.

3.3 How the specific inventory was established

The inventory was established by consolidating genre-bearing categories from existing resources, rather than by independently annotating a pooled collection with 74 newly devised labels. Source labels or the defined genre of a source corpus provide the specific category; a shared broad parent and a participant-facing English description then organise that category within AraGenre. The resulting inventory reflects resource availability and inherited granularity, not a systematic census of all Arabic genres. The distributed evaluation files do not include an excluded-category log or a source-document join from which the exact resource-selection history could be reconstructed.

The final labels expose this mixed granularity. Informative contains 20 occupational recruitment categories, 13 subjects of book descriptions, six news topics, and three encyclopaedic subjects. Learning contains eight student-writing topics and four textbook levels. Creative contains five dialectal song-lyric categories and two poetry categories. These counts are descriptive groupings of the re-

Broad branch	Genres	Segments	Share	Median	Organising rationale
Informative	42	14,243	50.92%	65	Reporting or describing information, including topic- and occupation-specific subtypes.
Creative	7	3,500	12.51%	6	Literary and expressive writing, including poetry and dialectal song lyrics.
Interactive	4	2,000	7.15%	13.5	Reviews, posts, and comments oriented towards participants or responses.
Learning	12	3,729	13.33%	91	Pedagogical texts and learner writing, differentiated by level or subject.
Legal	3	1,500	5.36%	27	Normative and institutional legal documents.
Religious	6	3,000	10.73%	24	Scriptural, devotional, and interpretative religious writing.
Total	74	27,972	100.00%	46	

Table 1: Released broad taxonomy and final-data composition. “Genres” counts specific labels in the released taxonomy. Median length is measured in whitespace-delimited units, not linguistically segmented Arabic tokens. The overall median is calculated over all segments. Appendix A gives the complete specific-label inventory.

leased identifiers and definitions, not an additional level used in scoring. Distinguishing two recruitment occupations or two news topics necessarily uses subject-matter information as well as recognition of the enclosing document type. We therefore interpret specific classification as recovery of the released source-grounded categories, not pure domain-independent genre recognition.

3.4 Definitions and annotation responsibilities

Initial English definition drafts were generated with ChatGPT 5 and reviewed, corrected, and manually post-edited by the organisers. This was definition preparation, not creation of new instance-level gold labels. The organisers’ construction account reports no new instance-level genre annotation or adjudication for the final collection, so no new instance-level inter-annotator agreement is claimed. The category-level grouping and definition review are separate curatorial decisions; they should not be presented as an independent semantic validation of every text.

The validation in Section 4.3 checks identifiers, label membership, parent consistency, and exact-text duplication. These checks provide reproducible structural quality control, but cannot establish whether an inherited category is the best semantic label for an individual segment. Appendix B.1 specifies the source information needed to make inheritance and category mapping fully traceable.

4 Data Collection and Benchmark Construction

4.1 Source resources and inherited labels

The contributing resource collection is described through the ArabicNLP.uk catalogue.¹ It includes Tarab (El-Haj, 2026), AraFinNews (El-Haj and Rayson, 2025), ArabJobs (El-Haj, 2025), the Multilingual Corpus of World’s Constitutions (MCWC; El-Haj and Ezzini, 2024), KALIMAT (El-Haj and Koulali, 2013), HABIBI (El-Haj, 2020), the Essex Arabic Summaries Corpus (EASC; El-Haj et al., 2010), the Arabic Dialects resource (El-Haj et al., 2018), DARES (El-Haj et al., 2024), and the Multi-Ling corpora (Li et al., 2013).

These publications document their own collection and quality-control procedures. Their annotations were produced for different purposes, so an original topic, occupation, dialect, or document-type category should not automatically be described as a manual genre annotation. AraGenre uses those categories or corpus-level identities as the basis for its task labels. The common taxonomy and descriptions do not retrospectively impose uniform annotation guidelines on the original resources.

The final collection is source-derived, in contrast to the primarily synthetic task-specific demonstrations. “Source-derived” does not certify that every Arabic sentence was originally authored in Arabic or untouched by upstream processing: for example, MCWC reports using machine translation to expand

¹<https://arabicnlp.uk/>

Split	Instances	Broad	Specific
Training	140	5	7
Development	110	6	6
Final	27,972	6	74

Table 2: Reported task splits. Final specific count refers to the released definition inventory. Training/development counts and label disjointness are taken from the task records.

language coverage (El-Haj and Ezzini, 2024). The evaluation snapshot contains no per-instance translation flag. Claims about the provenance of particular final segments must therefore remain separate from the general description of the contributing resources.

4.2 Released supervision and final evaluation

Table 2 reports the task split sizes. Participants received 140 labelled training examples covering seven specific genres from five broad branches. Development contained 110 examples covering six specific genres and all six branches. The final taxonomy introduced 74 defined specific genres absent as labelled classes from those releases. Names and English definitions were provided for the final classes, but no gold-labelled demonstrations of those classes were released for the final phase.

Several factors change together: the number and visibility of specific classes, amount of evaluation data, synthetic versus source-derived composition, source, topic, dialect, register, orthography, and text length. The development-to-final difference consequently measures performance under a combined change of conditions. It cannot identify the separate effect of unseen labels, nor establish that simply increasing the number of test instances makes the classification problem harder.

4.3 Final-data validation

We validated the final evaluation data using the released input texts, gold labels, and genre definitions. There are 27,972 unique input identifiers and 27,972 unique gold identifiers, with exact agreement between their identifier sets and no blank input text. All 74 defined specific labels occur in the gold data. Defined-class support ranges from eight to 500 instances, with 41 classes having exactly 500. Overall segment length ranges from one to 600 whitespace-delimited units, with median 46. Table 1 shows substantial differences in length and support across broad branches.

Exact string comparison identified 139 repeated-text groups containing 295 instances. Some identical strings occur with different specific labels, while their broad labels remain consistent. Because the supplied evaluation files do not contain the source-document metadata needed to determine whether these cases arise from source context, category inheritance, or other aspects of dataset construction, we report them descriptively rather than treating them as annotation errors. This observation further motivates retaining source provenance in future benchmark releases. Appendix B provides further details on the validation procedure.

4.4 Genre, source, and domain cues

The label inventory makes some confounding explicit: news topics, recruitment occupations, and lyric varieties are components of specific labels. Other shortcuts may come from corpus-specific boilerplate, named entities, formatting, register, and length. Input files contain only an identifier and text, not a source field, but withholding an explicit source identifier does not remove textual provenance cues. The identifiers are required for prediction alignment and are not intended as semantic features.

The validation describes dataset composition and exact-text repetition, but does not establish shortcut use by particular systems. Without source-document metadata and participant predictions, we cannot measure source reliance or report source-conditioned errors. A high score therefore supports recovery of the released categories under the available signals, not source-invariant understanding of communicative purpose. A stronger diagnostic would hold out sources while retaining the same specific labels across sources and controlling topic and length; merely holding out a corpus that uniquely defines a class would confound source shift with label shift. Appendix D describes the required controls.

5 Evaluation Protocol

5.1 Submission and ranking

Predictions were submitted as JSON through Codabench.² The task’s scoring procedure aligns predictions with gold instances by identifier. Every gold identifier is evaluated; missing predictions or missing broad/specific labels receive INVALID. Predictions for unknown identifiers are ignored, and

²<https://www.codabench.org/competitions/16356/>

the final prediction is retained when a submission repeats an identifier. These rules describe the competition protocol; they are not a new scoring implementation inferred from the evaluation files.

Macro F1, Weighted F1, and accuracy are calculated separately at the two levels. Macro F1 averages class-wise F1 equally, while Weighted F1 weights it by gold support. The scorer uses scikit-learn with undefined F1 values set to zero. All F1 values in this paper are on the 0–100 scale. The ranking score, termed *Hierarchical Macro F1* in the shared task, is

$$F_{\text{hier}} = \frac{F_{\text{macro}}^{\text{broad}} + F_{\text{macro}}^{\text{specific}}}{2}. \quad (1)$$

It is an equal-weight average of two independently evaluated tasks, not a path-aware hierarchical loss. A correct broad prediction receives broad-level credit even when the specific prediction is wrong, and inconsistent pairs are not directly penalised beyond their component errors.

5.2 Hierarchy consistency and evidence limits

A complementary consistency diagnostic measures the percentage of prediction pairs for which the specific label is defined and its parent equals the predicted broad label. This statistic is distinct from accuracy: a pair may be perfectly consistent yet wrong at both levels. Its formula and treatment of invalid or missing outputs are given in Appendix C.

Neither the aggregate results used for this overview nor the three-file data archive contains the participant prediction pairs required to calculate per-entry consistency. Gold labels cannot recover those pairs, and broad and specific Macro F1 values do not determine their joint behaviour. We therefore do not report per-entry consistency rates or treat the gold-path check as a substitute. Adding this diagnostic would not require retraining, only the original final submissions and the released parent mapping. The official ranking therefore remains separate from this proposed diagnostic.

6 Organiser Baselines

Two organiser baselines were provided. **Baseline 1** embeds Arabic inputs and English genre definitions with a multilingual sentence encoder and predicts through semantic similarity. This follows the sentence-embedding approach of Reimers and Gurevych (2019), although cross-lingual performance depends on the chosen multilingual encoder.

Baseline 2 uses an instruction-tuned multilingual language model with retrieved demonstrations and genre definitions in its prompt, following a retrieval-augmented formulation (Lewis et al., 2020). The starter kit provides loaders, evaluation code, and submission formatting.

The reported development Hierarchical Macro F1 scores are 46.58 for Baseline 1 and 38.99 for Baseline 2. Baseline 1’s official final score is 24.72. These are competition reference scores, not new baseline runs. The drop is consistent with the jointly changed evaluation conditions but cannot be assigned to one factor. Exact encoder/model identifiers, versions, prompts, and the frozen scorer are needed alongside the starter kit to reproduce the historical runs; the final JSON files alone do not supply those details.

7 Participating Systems

AraGenre received 46 registrations and 373 submissions. The official final leaderboard contains 17 ranked entries, while 11 system description papers document the main submitted approaches. We use *entries* for leaderboard rows because NAMAA-1/NAMAA-2 and Nasaq-1/Nasaq-2 share system-family descriptions; row count should not be interpreted automatically as a count of independent research groups. No unique-group total is reconstructed from usernames.

The descriptions identify three recurring directions: direct definition-guided LLM classification; cross-lingual retrieval between Arabic texts and English descriptions; and combinations of encoders or retrieval with LLM reranking, synthetic examples, pseudo-labels, calibration, or hierarchical post-processing. Tables 3 and 4 summarise the documented methods. Model names and implementation details in these summaries are attributed to the corresponding system descriptions, rather than independently reproduced experiments.

The six highest-ranked documented system families use an LLM at some stage, but the roles differ. Thakaa and NAMAA use LLMs for direct classification or adjudication; HP uses LLM pseudo-labels to train encoders; GHADAI, Nasaq, and Phoenix combine retrieval with LLM-based decisions. This is an observed association, not evidence that LLM use alone causes higher scores. SAGE and Mohanad Hendi use adapted embedding spaces, while WathiqTeam incorporates broad-first restriction and selective reranking. Northwestern provides a direct

System family	Main strategy	System summary
Thakaa (Alqaeri et al., 2026)	Definition-guided LLM judging	Uses a family-restricted LLM judge followed by agreement-gated correction. Two instruction models re-predict labels while viewing the 74-genre taxonomy, with a within-family attractor-drain procedure used to redistribute predictions towards under-predicted genres.
HP (Phong and Thin, 2026)	LLM pseudo-labelling + encoders	Uses definition-guided few-shot LLM prompting to pseudo-label the final data, then fine-tunes MARBERTv2 and ARBERTv2 on approximately 28,000 generated labels.
NAMAA (Barmandah et al., 2026)	Zero-shot LLM + self-consistency	Uses zero-shot DeepSeek classification using the official definitions, then applies three-sample self-consistency voting and disagreement resolution, including a Claude-assisted final pass.
GHADAI (Alsubhi and Kurdi, 2026)	Retrieval + LLM reranking	Combines a fine-tuned multilingual E5 bi-encoder for definition retrieval with Qwen2.5-32B reranking of shortlisted genres, using cyclic option rotation to reduce positional bias.
Nasaq (Altamimi et al., 2026)	Synthetic prototypes + selective LLM adjudication	Introduces SynAnchor, which generates Arabic genre prototypes from English definitions with GPT-4o, retrieves candidates using BGE-M3, and invokes GPT-5 selectively to adjudicate ambiguous cases.
Phoenix (Chi, 2026)	Multi-view retrieval + calibrated reranking	Combines multilingual retrieval with per-genre z-normalisation, then reranks candidates using calibrated Qwen3-8B log-probabilities, score fusion, sibling reranking, metric-aware decoding, and margin-gated candidate expansion.

Table 3: Documented participating systems, part I.

zero-shot retrieval reference.

Batch adaptation also matters to interpretation. For example, HP’s use of pseudo-labelled final inputs differs from a classifier applied independently to each unseen text. The overall task has unseen gold-labelled classes, but not every submission is inductive or free from task-specific training. Differences in adaptation, proprietary models, generated data, prompts, compute, ensembling, and post-processing are not controlled across entries.

8 Official Results and Analysis

Table 5 reproduces the official final leaderboard. Values are retained at their reported precision; averaging already rounded component scores can differ in the last displayed digit from a score computed before rounding. The official scores are reported unchanged.

8.1 Ranking and broad-specific differences

Thaka ranked first with Hierarchical Macro F1 of 73.52, followed by HP at 71.69 and NAMAA-1 at 70.13. The top three span 3.39 points. HP had the highest Broad Macro F1, 90.00 versus Thaka’s 89.79, whereas Thaka led on Specific Macro F1, 57.25 versus HP’s 53.39. Its stronger specific classification therefore produced the highest combined score under Equation 1.

GHADAI, Nasaq-1, and phoenix form the next group, scoring 58.00–60.50, followed by SAGE at 52.78. Every final entry exceeds the organiser baseline of 24.72. These numerical differences describe the submitted configurations; they are not significance tests or estimates of architecture effects.

Across the 17 entries, mean Broad Macro F1 is 65.09 and mean Specific Macro F1 is 32.38, a 32.71-point gap. Every entry performs better on the broad task. This is compatible with the much smaller upper-level inventory and largely shared broad labels, alongside numerous unseen specific classes. It does not establish that the broad branches are semantically disjoint, or that finer communicative distinctions alone explain the gap: topic, source, variety, class count, and inherited label ambiguity also differ between the two tasks.

8.2 Development-to-final change

For 13 final entries whose usernames were matched to development submissions in the task records, the reported mean best development score was 93.44 and the mean final score was 52.61, a difference of approximately 40.8 points. Baseline 1 likewise falls from 46.58 to 24.72. These comparisons describe the joint change of conditions in Section 4.2. Best development scores also reflect repeated selection on the development benchmark, whereas the final

System family	Main strategy	System summary
SAGE (Mansour et al., 2026)	Fine-tuned sentence encoders	Fine-tunes sentence encoders for genre similarity and learns a residual adapter to refine definition embeddings, targeting the representational gap between English descriptions and Arabic texts.
Mohannad Hendi (Hendi, 2026)	Contrastive definition retrieval	Fine-tunes multilingual E5-large contrastively on Arabic texts paired with five deterministic views of English definitions, then performs nearest-definition retrieval in the shared embedding space.
WathiqTeam (Bostaty et al., 2026)	Broad-first hybrid retrieval	Combines multilingual E5, Arabic prototype retrieval, and BM25 using class-balanced pooling, reciprocal-rank fusion, hierarchical gating, and selective Qwen3 reranking for uncertain cases.
Rabee Al-Qasem (Al-Qasem, 2026)	Embeddings + hierarchical supervision	Combines multilingual MPNet and genre-definition representations with a Conformer projection module, auxiliary broad-genre supervision, synthetic augmentation, and seed-based ensembling.
Northwestern (Shlkamy and Zaghoulani, 2026)	Zero-shot dense retrieval	Encodes Arabic texts and English genre definitions with multilingual BGE-M3 and assigns labels by cosine similarity, without task-specific fine-tuning or additional supervised training.

Table 4: Documented participating systems, part II.

setting has different labels and texts.

The means do not provide a correlation or establish how well development rankings predict final rankings. The defensible conclusion is that high performance on the small controlled development set does not guarantee comparably high performance in the combined final setting. Separating label novelty from source and linguistic shift requires controlled data cells, not reinterpretation of the aggregate leaderboard. Appendix D specifies a possible design, explicitly as future work.

8.3 What the system comparison supports

The documented high-ranked systems all use label semantics, through prompts, embeddings, synthetic prototypes, or pseudo-labelled supervision. This is consistent with the task’s definition-guided design. It does not isolate whether their gains come from the definitions, model scale, retrieval, generated data, calibration, or additional inference. A supervised classifier restricted to the released class IDs also lacks a direct way to output all final labels unless its output mechanism or training data are adapted.

Retrieval followed by reranking is a recurring practical design: a first stage shortlists candidates and a second stage compares more difficult alternatives. The submissions show several implementations, rather than a controlled demonstration that this design is universally superior. Likewise, the broad-specific gap motivates investigation of hierarchy-aware decoding, but does not prove that current systems underuse the hierarchy. Gold-consistent predictions, source-conditioned er-

rors, within-family confusions, and computational cost require additional evidence beyond component Macro F1.

9 Lessons for Future Editions

A useful priority for future editions is a versioned data manifest linking each segment to its source document, original category, label derivation, and any transformation applied during benchmark construction. The full label inventory in Appendix A documents the current task taxonomy, while a more detailed provenance manifest would make the relationship between the consolidated benchmark and its contributing resources easier to trace. Future release checks could also include automated verification of label inventories, broad-specific mappings, identifier uniqueness, and exact text duplication before the benchmark is frozen. These measures would further strengthen the transparency and reproducibility of subsequent editions.

A second priority is richer evaluation evidence. Retaining final prediction files would allow hierarchy-consistency rates, per-class confusion analysis, and stratification by length. Source and dialect diagnostics additionally require reliable metadata. Standardised system fact sheets should record model versions, training and external data, synthetic or pseudo-labelled examples, access to the full final batch, prompts, compute, and post-processing. Participation reports should distinguish registrations, active entrants, ranked entries, independent groups, and system papers.

Rank	Leaderboard entry	Hierarchical	Specific	Broad
1	Thaka (Alqaeri et al., 2026)	73.52	57.25	89.79
2	HP (Phong and Thin, 2026)	71.69	53.39	90.00
3	NAMAA-1 (Barmandah et al., 2026)	70.13	51.14	89.13
4	GHADAI (Alsubhi and Kurdi, 2026)	60.50	42.97	78.03
5	Nasaq-1 (Altamimi et al., 2026)	60.06	42.64	77.48
6	phoenix (Chi, 2026)	58.00	41.09	74.92
7	SAGE (Mansour et al., 2026)	52.78	35.05	70.51
8	Mohannad Hendi (Hendi, 2026)	45.60	29.95	61.25
9	WathiqTeam (Bostaty et al., 2026)	45.59	28.69	62.50
10	rsonbol	44.21	27.24	61.19
11	Rabee Al-Qasem (Al-Qasem, 2026)	43.99	25.11	62.88
12	Nasaq-2 (Altamimi et al., 2026)	35.80	17.14	54.47
13	NAMAA-2 (Barmandah et al., 2026)	35.55	21.29	49.81
14	NasSem	33.67	19.30	48.04
15	Northwestern (Shlkamy and Zaghouani, 2026)	33.02	20.08	45.96
16	GenreMinds	32.86	19.23	46.50
17	morooj_6	31.52	18.94	44.10
–	AraGenre baseline	24.72	13.58	35.85

Table 5: Official final leaderboard, retained as reported, with Macro F1 on a 0–100 scale. Rows are ranked entries, not necessarily distinct research groups. The two NAMAA and two Nasaq entries remain separate.

Finally, future evaluation should cross label visibility with source shift while maintaining comparable class support and linguistic composition. Source-held-out and topic-controlled subsets can test whether performance survives changes in provenance, but only when the same target categories remain identifiable in the comparison. Appendix D states the necessary controls and the analyses that the present evidence cannot support.

Limitations

The resource-driven taxonomy mixes communicative function, institutional setting, document type, topic, occupation, and variety. Its specific labels are unevenly distributed across broad branches, and support and segment lengths vary. Macro F1 reduces dominance by frequent instances but does not remove differences in granularity, structural imbalance, or source dependence.

Gold labels are inherited from heterogeneous resources rather than produced through a single instance-level genre-annotation protocol. The evaluation data supplied for the present analysis do not encode record-level provenance, translation status, dialect, original source labels, or the selection decisions associated with individual instances. Our validation therefore focuses on the structural properties that can be established directly from the released test data, gold labels, and genre definitions,

including identifier correspondence, label inventories, broad-specific mappings, and exact text duplication. These checks assess the internal structure of the benchmark and should not be interpreted as a semantic agreement study or as a reassessment of the original source annotations.

Several factors change simultaneously between development and final evaluation. System configurations also differ in data, models, prompts, batch adaptation, and computational resources. Complete prediction files are not available for the analysis presented here, so no per-entry consistency rates, significance tests, source-specific errors, or causal architecture comparisons are reported. These limitations constrain interpretation without invalidating the descriptive account of the task and its official results.

Ethics and Resource Use

AraGenre consolidates resources with heterogeneous collection and access conditions. Public availability alone does not establish a uniform licence, permission to redistribute text, or absence of personal information. Downstream use should follow each source’s terms and document source-specific privacy and copyright considerations. The validation reported here uses aggregate counts and structural properties rather than reproducing Arabic passages from the evaluation data. This analysis

does not create new semantic annotations or constitute a new human-subject study. The released taxonomy is a benchmark organisation of texts, not a judgement about the legitimacy or value of the represented linguistic varieties, occupations, or discourse traditions.

10 Conclusion

AraGenre 2026 provides a definition-guided Arabic classification task with six broad branches, 74 defined specific genres, and 27,972 final segments. Its main challenge is transfer from limited released supervision to an expanded label inventory and substantially different evaluation material. The official leaderboard contains 17 ranked entries, with 11 accompanying system papers; Thaka achieved the highest Hierarchical Macro F1 at 73.52. Broad classification consistently exceeded specific classification, and documented systems frequently combined label descriptions with multilingual retrieval, LLM inference, or task adaptation.

AraGenre should therefore be interpreted as evaluating inherited source-grounded categories under combined distribution shift, rather than isolated unseen-label transfer or source-invariant genre understanding. The complete inventory and validation analysis clarify both the available evidence and the provenance information that would strengthen future editions. Future editions should retain prediction artefacts, report hierarchy consistency, document record-level provenance, and separate label shift from source and linguistic shift through controlled evaluations.

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Appendix

A Complete Final Label Inventory

Tables 6 and 7 enumerate all 74 identifiers in `test_genre_definitions.json`, preserving their released spelling and broad parent. Counts use exact label matching in `test_gold.json`; medians use whitespace-delimited lengths of the corresponding input texts. These are observed support counts, not reconstructed sampling quotas. Every defined class is represented, the smallest has eight instances, and 41 classes have 500 instances.

The inventory documents the taxonomy represented in the released genre-definition file. The available evaluation files do not provide a verified record-level mapping to source corpus, original source label, dialect, or translation status; these provenance boundaries are discussed in Appendix B.1.

B Provenance and Data Validation

B.1 File scope and provenance boundary

The inspected archive contains exactly three JSON arrays. Input records have `id` and `text`; gold records have `id`, `broad_genre`, and `specific_genre`; definition records have broad/specific identifiers and their English definitions. It contains no training or development data, participant predictions, scoring implementation, per-instance source corpus, source document ID, original source label, dialect annotation, or translation indicator.

Three kinds of evidence are therefore distinguished. The task records supply the split design, inheritance account, participation figures, and historical scores. The source publications describe their own corpora. The validation independently establishes structural properties of the supplied final files. A source publication or a plausible match between a label name and a corpus is not evidence of the exact source record for a particular benchmark instance.

To close this provenance gap, a versioned manifest should contain, for each segment, its benchmark ID, source resource and version, source document ID and segment offsets, original category, derivation type (direct label retention, corpus-level assignment, or category mapping), specific and broad task labels, and any translation or text transformation. A separate mapping log should identify the category-level decisions, their rationale, and any re-

view or resolution of ambiguous mappings. These are documentation requirements, not claims that missing fields have been recovered. The inventory CSV explicitly marks source corpus and original source label as not recorded.

B.2 Validation procedure and identity checks

To validate the final evaluation data, we performed a series of consistency and integrity checks directly on the released test, gold-label, and genre-definition files. We first verified that all required fields were present, that instance identifiers were unique, and that the test and gold files contained exactly the same set of identifiers. We then reconstructed the specific-to-broad genre hierarchy from the released genre definitions and verified the gold-labelled instances against this mapping to assess the structural consistency of the taxonomy and broad-specific assignments.

We additionally examined basic properties of the evaluation texts and labels. Text length was measured using whitespace-delimited units, without Arabic-specific segmentation or orthographic normalisation. To identify exact duplication, texts were compared in their released form, without case folding, trimming, diacritic removal, normalisation, or approximate matching. Consequently, an “exact duplicate” refers strictly to two or more instances containing identical text strings; it does not include near-duplicates or texts that are semantically similar but lexically different. These checks were diagnostic only: no test instances, identifiers, or gold labels were altered as a result of the analysis.

The validation identified 27,972 test instances and 27,972 corresponding gold records, with unique identifiers and exact agreement between the two identifier sets. No blank test texts were found. All 74 genres included in the released genre-definition inventory occur in the evaluation data. We additionally verified the broad-specific assignments against the parent mapping represented in the released taxonomy. These checks establish the structural organisation of the benchmark; they do not assess the semantic correctness of the inherited genre labels or the hierarchical consistency of participant predictions.

We also examined the evaluation data for exact text repetition. Identical text strings occur across a small number of records, including cases in which the same text is associated with different specific genres. We retain these records when describing and analysing the official benchmark because they

Released specific identifier	Broad	Count	Median
admin_and_secretarial_jobs	Informative	500	55
arts_book_description	Informative	146	64
automotive_and_mechanics_jobs	Informative	420	37.5
biographies_memoirs_book_description	Informative	234	67
cleaning_services_jobs	Informative	272	26.5
culture_encyclopedia	Informative	500	52
culture_news	Informative	500	424.5
customer_service_jobs	Informative	386	46
drivers_and_delivery_jobs	Informative	140	29
economics_business_book_description	Informative	395	62
economy_news	Informative	500	343.5
education_jobs	Informative	93	35
engineering_jobs	Informative	353	30
family_children_book_description	Informative	324	63
finance_and_accounting_jobs	Informative	500	55.5
health_and_beauty_jobs	Informative	212	20
healthcare_jobs	Informative	245	29
history_encyclopedia	Informative	500	40
history_geography_book_description	Informative	122	62
human_resources_jobs	Informative	264	71.5
information_technology_jobs	Informative	67	75
international_news	Informative	500	322
islamic_book_description	Informative	471	64
journalism_media_book_description	Informative	259	62
law_and_legal_services_jobs	Informative	37	43
literature_fiction_book_description	Informative	500	64
local_news	Informative	500	335.5
manufacturing_and_retail_jobs	Informative	228	65
marketing_jobs	Informative	415	102
math_encyclopedia	Informative	500	51
media_and_design_jobs	Informative	206	54.5
philosophy_book_description	Informative	500	66
political_book_description	Informative	389	64
religion_news	Informative	500	552.5
sales_jobs	Informative	500	55
science_nature_book_description	Informative	355	66
security_jobs	Informative	133	63
sharia_law_book_description	Informative	19	63
sports_entertainment_book_description	Informative	57	67
sports_news	Informative	500	270.5
technicians_and_craftsmen_jobs	Informative	500	24.5
tourism_and_restaurants_jobs	Informative	500	26
Defined-label subtotal		14,242	

Table 6: Complete final inventory, part I: 42 Informative labels. Median is the number of whitespace-delimited units. Exact released identifiers are retained; line breaks do not change spelling.

Released specific identifier	Broad	Count	Median
classical_poetry	Creative	500	5
egyptian_song_lyrics	Creative	500	7
gulf_song_lyrics	Creative	500	6
iraqi_song_lyrics	Creative	500	6
levantine_song_lyrics	Creative	500	6
msa_poetry	Creative	500	5
sudanese_song_lyrics	Creative	500	6
app_reviews	Interactive	500	10
book_reviews	Interactive	500	42
twitter_posts	Interactive	500	12
youtube_comments	Interactive	500	11
communication_student_writing	Learning	115	506
culture_student_writing	Learning	411	148
early_education_textbooks	Learning	457	42
high_school_textbooks	Learning	500	54
intermediate_textbooks	Learning	500	51
national_development_student_writing	Learning	8	159.5
schoolwork_student_writing	Learning	500	150
social_media_student_writing	Learning	127	152
sports_student_writing	Learning	500	159
technology_student_writing	Learning	80	504
upper_elementary_textbooks	Learning	500	46
values_student_writing	Learning	31	139
constitutions	Legal	500	14
contracts	Legal	500	40
resolutions	Legal	500	33
Biblical Scripture	Religious	500	9
Biblical Commentary	Religious	500	34
coptic_devotional_literature	Religious	500	30
hadith	Religious	500	56
quran	Religious	500	11
tafsir	Religious	500	26.5
Defined-label subtotal		13,729	

Table 7: Complete final inventory, part II: the 32 labels in the other five branches. Together with Table 6, this covers all 74 specific genres in the released definition inventory.

formed part of the evaluation data used for the shared task. Rather than retrospectively modifying the benchmark, we report these cases transparently as a property of the released evaluation set and discuss their implications for fine-grained genre evaluation.

For reproducibility and data-version identification, Table 8 reports the SHA-256 checksums of the three files used for these validation checks. This makes the precise version of the evaluation data examined here independently identifiable without requiring any modification to the original benchmark.

B.3 Exact-text repetition

There are 27,816 distinct exact text strings among the 27,972 evaluation instances. Exact comparison identifies 139 repeated-text groups containing 295 instances: 128 groups of size two, seven of size three, two of size four, and two of size five. Some identical strings are associated with different specific genres, while their broad labels remain consistent.

These cases are reported as a structural property of the released evaluation data rather than interpreted as annotation errors. The available files do not contain the source-document metadata needed to determine whether the repetitions reflect source context, category inheritance, repeated material across contributing resources, or other aspects of dataset construction. Future releases would benefit from retaining source-document provenance so that such cases can be interpreted directly.

B.4 Composition is not an error analysis

The branch counts and length distributions in Table 1 are descriptive properties of the gold collection. They do not establish that a system uses length as a shortcut. Likewise, genre-name groupings reveal mixed label granularity but do not measure reliance on a particular source. No dialect-specific prevalence is estimated from label names, and no confusion matrix is reconstructed without predictions. The validation is deliberately limited to properties supported by the supplied files.

C Prediction-Consistency Diagnostic

To assess whether systems preserve the intended hierarchy between broad and specific genres, we define a prediction-consistency diagnostic. Let N denote the number of gold identifiers and let p map

each of the 74 defined specific genres to its corresponding broad genre. After aligning system predictions with the gold data by identifier, the unconditional hierarchical consistency rate is defined as

$$C = \frac{100}{N} \sum_{i=1}^N \mathbf{1}[\hat{s}_i \in \mathcal{S} \wedge \hat{b}_i = p(\hat{s}_i)], \quad (2)$$

where $\hat{s}_i$ and $\hat{b}_i$ denote the predicted specific and broad genres, respectively, and $\mathcal{S}$ is the inventory of defined specific genres. Missing predictions, invalid broad labels, and specific predictions outside the permitted inventory contribute zero. The denominator remains all N evaluation instances, preventing consistency from being artificially increased by omitting difficult cases. A complementary valid-pair coverage measure can report the proportion of instances for which both predictions belong to their respective label inventories. Consistency among valid prediction pairs may also be reported, provided that its denominator and coverage are stated explicitly.

Hierarchical consistency should be interpreted separately from classification correctness. A prediction can be internally consistent while being wrong at both levels, for example when an incorrect specific genre is returned together with its corresponding broad parent. Conversely, a correct broad prediction can be paired with an incompatible specific prediction. The consistency rate in Equation 2 therefore complements the Broad and Specific Macro F1 measures rather than replacing either of them.

The evaluation data supplied for the present analysis contain the test instances, gold labels, and genre definitions, but not the instance-level predictions submitted by participating systems. We can therefore verify the structural consistency of the gold broad-specific paths, as described in Appendix B.2, but cannot retrospectively calculate C for individual participating systems. For this reason, we do not report per-system consistency values.

This distinction is important when interpreting hierarchical modelling strategies. Agreement between a predicted specific genre and its parent does not by itself demonstrate that hierarchical inference improves classification. A hard broad-first constraint may propagate an incorrect broad decision to the specific level, whereas independent prediction may occasionally preserve a correct label at one level despite disagreement at the other. Establishing whether hierarchical constraints improve performance would therefore require instance-level

Evaluation file	SHA-256
test.json	e81b1cab662bfd56b5cfe27e9a7fb2a0 f5e262ad4c66b671644f7e426c5d2be7
test_gold.json	3ba033939de472aad63a3b0c49ab2723 54a49cb89f8438bfa9f68a8b79c3be4b
test_genre_definitions.json	a0eb71d6df773a660b856b152ea7410f fa6ae5cea4658918c98c72af4952ff90

Table 8: SHA-256 checksums of the evaluation files used for the validation reported in this paper.

system outputs or controlled comparisons between otherwise matched inference strategies. We recommend reporting hierarchical consistency alongside the existing evaluation measures in future editions of the task.

D Controlled Evaluation Design for Future Editions

A useful next design crosses two factors: whether specific labels are represented in released supervision, and whether test sources differ from the sources represented during development. This produces four cells: seen labels/source matched, unseen labels/source matched, seen labels/source shifted, and unseen labels/source shifted. All four are proposed diagnostics; no results for them are claimed in this paper.

To isolate label visibility, the seen and unseen category sets should be matched on broad branch, support, label granularity, and difficulty proxies, preferably rotating category assignments across replicates. Both cells should use source-derived text rather than conflating label novelty with a synthetic-to-natural transition. To isolate source shift, the same target categories must occur in several sources, with whole documents and exact/near-duplicate groups kept together. Otherwise a source-held-out test may simply remove all evidence for a class.

Class counts, support, length, dialect, and topic distributions should be held comparable or explicitly stratified. A complementary source-prediction probe can establish whether provenance is recoverable, but does not alone prove that a genre classifier relies on it. Evidence of reliance requires suitable interventions or performance changes under source-controlled evaluation. Topic-controlled comparisons should also distinguish fine labels that explicitly encode topic from document-type distinctions intended to generalise across topics.

Evaluation should retain the current broad, specific, and combined metrics, add valid-output cov-

erage and Equation 2, and publish per-class and per-source results where support permits. If confidence intervals are estimated, resampling should respect source documents or duplicate groups rather than assume every segment is independent. These diagnostics require additional provenance and prediction artefacts; they are not recoverable from the current aggregate leaderboard and final-only JSON snapshot.